

الرياضيات المتقطعة الهندسية

رياضيات متقطعة

محاضرة أولى
مرحلة ثانية الذكاء الاصطناعي

المحاضرة الأولى

مرحلة الثالثة

قسم الذكاء الاصطناعي

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■ Lecture: Set Theory and Logical Relations (Discrete Mathematics – Geometric Approach)

1. Introduction

Discrete Mathematics deals with *discrete structures* such as sets, relations, logic, and graphs.

Two fundamental topics are:

- **Set Theory** (how we group and operate on elements)
- **Logical Relations** (how we connect and reason with statements).

They form the foundation for algorithms, computer science, and artificial intelligence.

2. Set Theory

2.1 Definition

A **set** is a collection of distinct, unordered objects, called *elements*.

- Example:

$$A = \{1, 2, 3, 4\}$$

Here, $2 \in A$ (2 belongs to A), but $5 \notin A$.

2.2 Ways to Define a Set

1. **Roster Method** (list elements):

$$B = \{a, b, c, d\}$$

2. **Set-builder Notation** (describe rule):

$$C = \{x \mid x \text{ is an even number less than } 10\}$$

$$\rightarrow C = \{2, 4, 6, 8\}$$

2.3 Types of Sets

- **Empty set:** $\emptyset = \{\}$
- **Subset:** $A \subseteq B$ means every element of A is in B .
 - Example: If $B = \{1, 2, 3, 4, 5\}$ and $A = \{2, 4\}$, then $A \subseteq B$.
- **Universal set:** contains all objects under discussion.
- **Equal sets:** $X = Y$ if they have the same elements.

2.4 Set Operations

- **Union ($\cup$)** – "OR"

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

$$\text{Example: } A = \{1, 2, 3\}, B = \{3, 4, 5\} \rightarrow A \cup B = \{1, 2, 3, 4, 5\}$$

- Intersection ($\cap$) - "AND"

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Example: Same $A, B \rightarrow A \cap B = \{3\}$

- Difference ($-$)

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

Example: $A = B = \{1, 2\}, B - A = \{4, 5\}$

- Complement (A')

If $U = \{1, 2, 3, 4, 5\}$ is the universal set and $A = \{1, 2\}$, then $A' = \{3, 4, 5\}$.

3. Logical Relations

3.1 Statements (Propositions)

A **statement** is a sentence that is either *true* (T) or *false* (F), but not both.

- Example:
 - "5 is an even number" $\rightarrow$ False
 - "Paris is the capital of France" $\rightarrow$ True

3.2 Logical Connectives

1. **Negation** ($\neg p$) $\rightarrow$ reverses truth value

Example: If p : "It is raining" $\rightarrow$ False,
then $\neg p$: "It is not raining" $\rightarrow$ True.

2. **Conjunction** ($p \wedge q$) - "AND"

True only if both are true.

Example:

- p : "I study"
- q : "I pass"
- $p \wedge q$: "I study AND I pass"

True

3. **Disjunction** ($p \vee q$) - "OR"

True if at least one is true.

Example: "I will go to the park OR to the cinema."

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4. Implication ($p \rightarrow q$) – "If...then..."

False only when p is true and q is false.

Example:

- p : "It rains"
- q : "The ground is wet"
- Statement: "If it rains, then the ground is wet."

5. Biconditional ($p \leftrightarrow q$) – "If and only if"

True when both have the same truth value.

Example: "You can enter the room if and only if you have a key."

3.3 Truth Table Example

p	q	p'	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

4. Connection Between Sets and Logic

- Union $\leftrightarrow$ OR
Example: "x is in A OR B"
- Intersection $\leftrightarrow$ AND
Example: "x is in A AND B"
- Complement $\leftrightarrow$ NOT
Example: "x is NOT in A"
- Difference $\leftrightarrow$ AND with NOT
 $A - B = A \cap B'$

5. Applications

- Programming: if, while, logical conditions.
- Databases: SQL operations (UNION, INTERSECT, MINUS).
- Artificial Intelligence: reasoning and rule-based systems.
- Geometry & Graphics: classification of objects into sets (e.g., visible vs. hidden edges)

Summary:

Set theory gives us a language to describe collections, while logic provides the rules to connect and reason about them. Together, they are the foundation of discrete mathematics and computer science.

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